

Road Warrior

EVENT-DRIVEN TRIP MANAGEMENT DASHBOARD

Team - 4



A large, stylized leaf graphic in a golden-brown color, positioned in the upper right quadrant of the slide. The leaf has a prominent central vein and several smaller veins branching out, giving it a detailed but simplified appearance. It overlaps with the background and the text area.

Problem, Scope, and Constraints



What Problem Are We Solving?

Unified dashboard

View and manage all existing travel reservations in one place across web and mobile

Reservation ingestion

Auto load reservations from agency airline, hotel, and car rental systems, with manual fallback entry

Trip management

Group reservations into logical trips with search/filter support and archive completed trips

Real-time experience

Provide live dashboard updates, notifications, and social trip sharing

Business visibility

Highlight preferred partner vendors while keeping new booking out of scope

Quality Drivers

Scalability

Support growth from 10k to 100k active users

Responsiveness

Rich UI across web and mobile with fast interactions

Elasticity

Absorb seasonal traffic spikes efficiently

Availability

Travelers must access current trip details globally

Data freshness

Updates such as delays or cancellations should appear quickly



Architecture Characteristics Worksheet

System/Project: Road Warrior Trip Managment

Domain/Quantum: _____

Architect/Team: Team 4

Date:03-05-2026 Next Review: _____

Top 3 Driving Characteristics



Scalability



Responsiveness



Elasticity



Availability



Data consistency



Instructions

- Identify no more than 7 driving characteristics.
- Pick the top 3 characteristics (in any order).
- Implicit characteristics can become driving characteristics if they are *critical* concerns.
- Add additional characteristics identified that weren't deemed as important as the list of 7 to the *Others Considered* list.
- Definitions are on the following pages

Implicit Characteristics

Performance

Security

Interoperability

observability

Others Considered

Common Architecture Characteristics

performance

data integrity

deployability

responsiveness

data consistency

testability

availability

adaptability

abstraction

fault tolerance

extensibility

workflow

scalability

interoperability

configurability

elasticity

concurrency

recoverability

Composite Architecture Characteristics

agility → maintainability, testability, deployability

reliability → availability, testability, data integrity,
data consistency, fault tolerance

^a denotes characteristics that are related; some systems
^b only need one of these, other systems may need both



Assumptions

Agency systems can publish events or expose efficient delta APIs

Authentication is handled by the agency identity provider

Trip grouping uses dates, locations, and travel sequence heuristics

Initial release focuses on English and UTC-based handling

Favored vendor status is provided by the agency interface layer

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Solution Strategy



Chosen Architecture Style and Rationale

Event-Driven Microservices

Allows independent scaling and asynchronous processing for reservations, notifications, and trip computations.

Modular Backend-For-Frontend

Simplifies client integration by providing a cohesive API surface, centralizing authentication and caching.

Scalability and Simplicity Balance

Combines microservices scalability with operational simplicity through streamlined client interactions.

Considered Alternatives

Synchronous monolith and serverless approaches were rejected due to scalability and predictability issues.

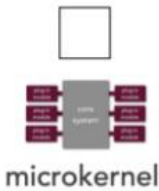
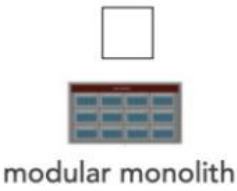
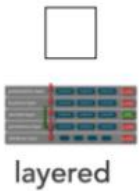
Architecture Styles Worksheet

System/Project: Road Warrior Trip Managment

Architect/Team: Team 4

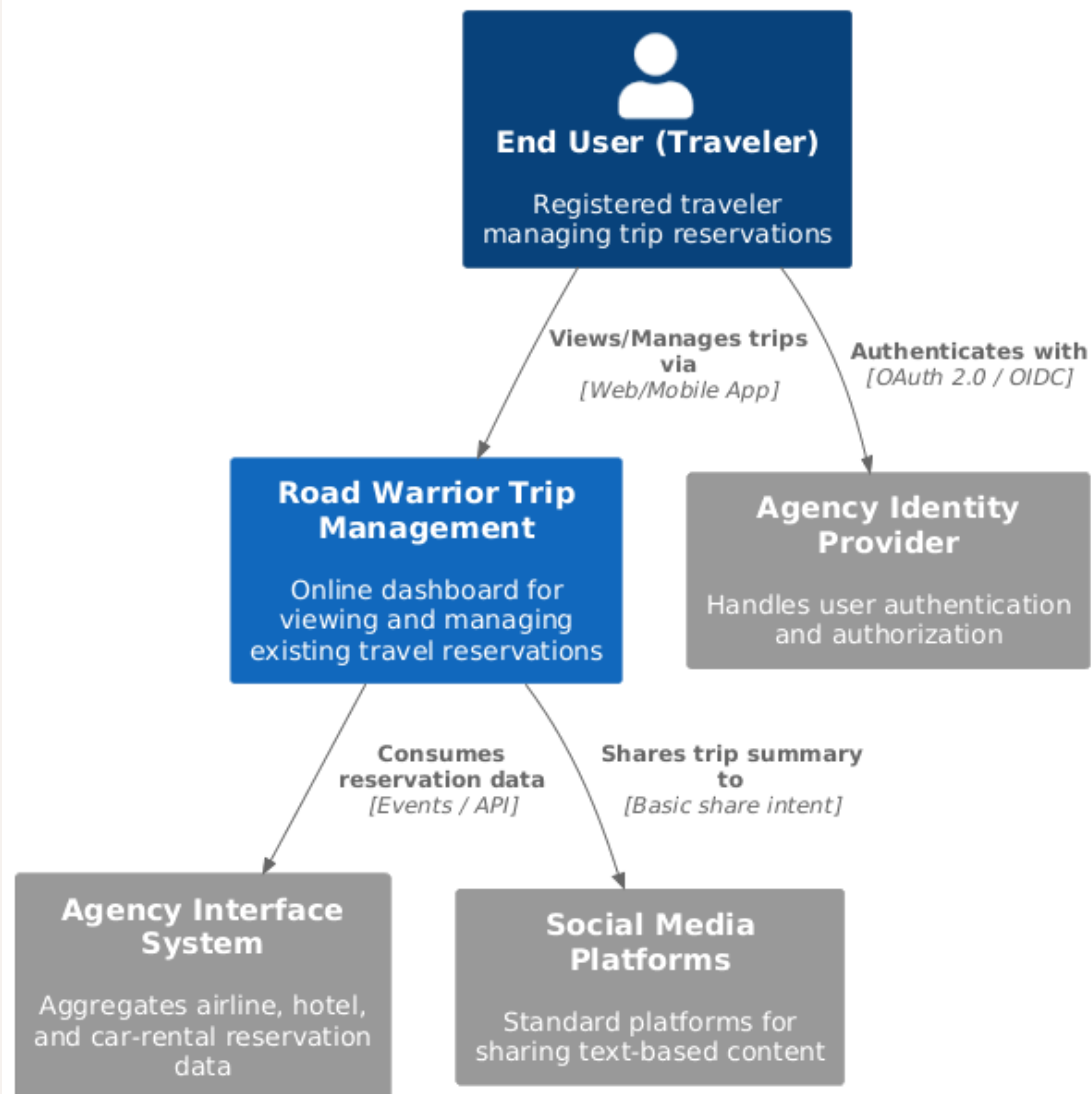
Date: 03-05-2026

Selected Architecture(s):



partitioning	technical	domain	domain	domain	domain	technical	technical	technical
cost	\$	\$	\$	\$\$\$\$\$	\$\$	\$\$\$\$	\$\$\$	\$\$\$\$
maintainability	★	★★	★★★	★★★★★	★★★★★	★	★★★	★★★★
testability	★★	★★	★★★	★★★★★	★★★★★	★	★★	★
deployability	★	★★	★★★	★★★★★	★★★★★	★	★★★	★★★★
simplicity	★★★★★	★★★★★	★★★★★	★	★★★	★	★★	★
scalability	★	★	★	★★★★★	★★★	★★★★	★★★★★	★★★★★
elasticity	★	★	★	★★★★	★★	★★★★	★★★	★★★★★
responsiveness	★★★	★★★	★★★	★★	★★★	★★	★★★★★	★★★★★
fault-tolerance	★	★	★	★★★★★	★★★★★	★★★★	★★★★★	★★★★
evolvability	★	★	★★★	★★★★★	★★★★★	★	★★★★★	★★★
abstraction	★	★	★★★	★	★	★★★★★	★★★★	★
interoperability	★	★	★★★	★★★★	★★	★★★★★	★★★	★★

System Context - Road Warrior Trip Management



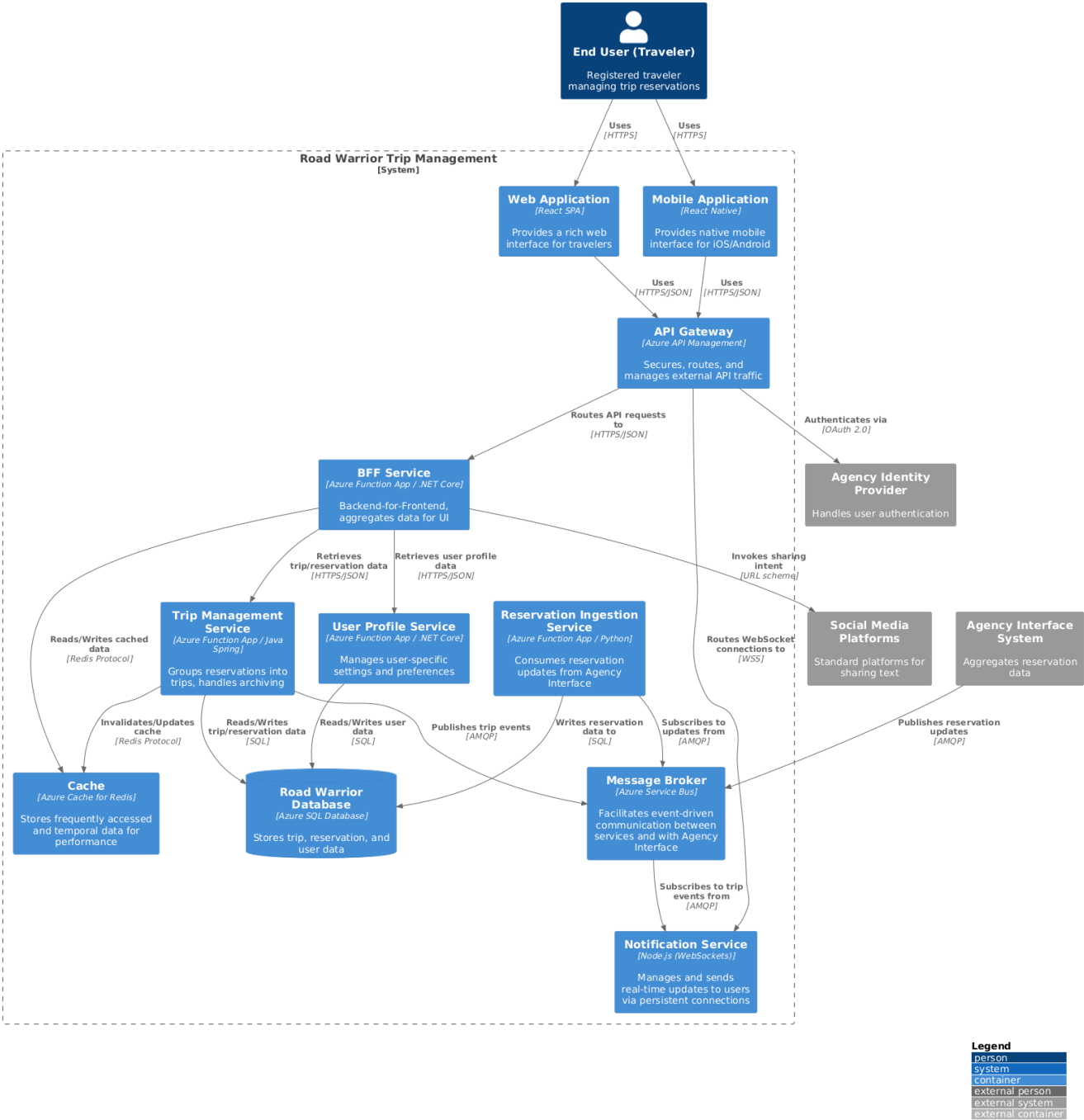
Legend

person
system
external person
external system

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Logical and Physical Architecture

Container Diagram - Road Warrior Trip Management



Logical Architecture

Web and mobile clients consume APIs through API Management and BFF

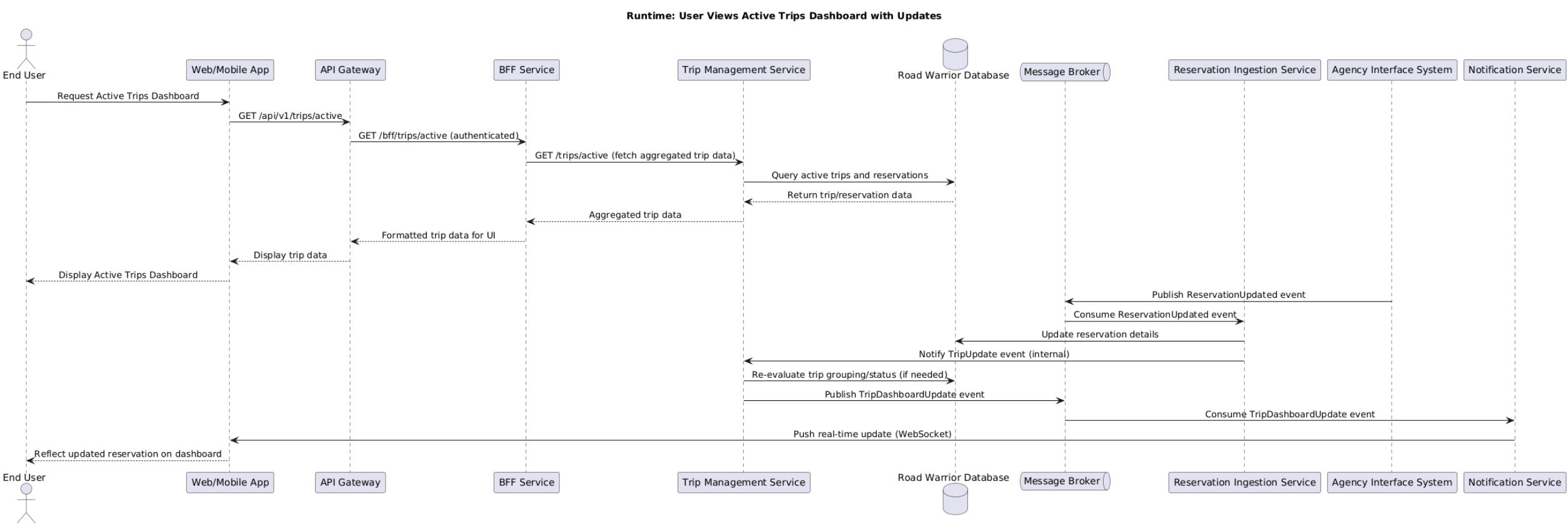
Reservation Ingestion Service processes agency-system updates

Trip Management Service groups reservations into trips

Notification Service pushes critical updates to clients

Azure SQL stores core data, Redis improves read performance, Service Bus carries events





Runtime Flow

Agency system publishes a reservation update event

Reservation Ingestion Service validates and persists the change

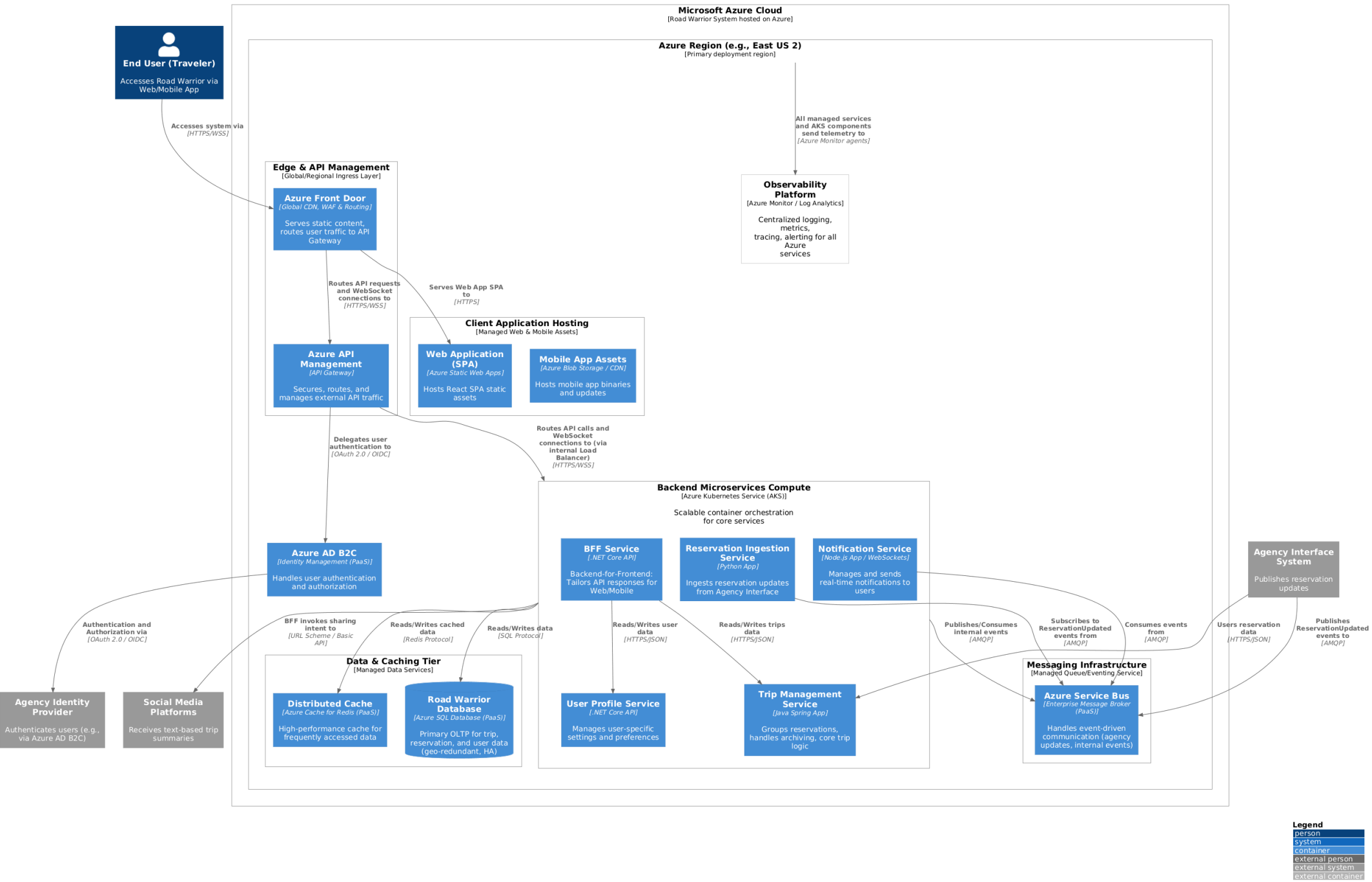
Trip Management updates trip grouping if needed

Notification Service pushes delay or cancellation changes to active clients

Dashboard reflects the latest state without user polling



Deployment View - Road Warrior Trip Management



Deployment Posture

Cloud-native deployment on Azure managed services

AKS hosts stateless backend services with horizontal auto-scaling

Azure SQL, Redis, and Service Bus provide managed data and messaging capabilities

Front Door and CDN improve global performance and edge protection

Observability supports distributed tracing, metrics, and alerting



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Decisions, Risks, and Concerns

Key Architecture Decision Records and Data Strategy

Microservices Architecture (ADR-1)

For independent scaling, fault isolation, and maintainability; tradeoff is distributed-system and operational complexity.

Event-Driven Data Ingestion (ADR-2 and ADR3)

Data ingestion uses event-driven approach for near real-time updates, fallback polling, retries, and failure handling via queues.

Managed Cloud Deployment (ADR-4)

To improve elasticity, availability, and operational efficiency; tradeoff is cloud cost control and vendor dependence.

Primary Data Store Choices (ADR-5)

Azure SQL is the primary operational store because trips, reservations, and user data are strongly relational, Redis improves read performance and Service Bus carries events.

Automated Trip Grouping Logic (ADR-6)

Reservations are grouped automatically by deterministic date and location rules while allowing manual user overrides for flexibility.





Key Risks and Concerns

Integration Dependency

The platform depends on the agency integration hub for reservation updates; mitigated through asynchronous messaging, retries, and polling fallback where needed.

Scalability and Burst

Seasonal demand spikes can overload synchronous paths, mitigated through stateless services, horizontal scaling, asynchronous workers, and queue-based buffering.

Data Consistency

Duplicate or delayed events can create inconsistent trip views, mitigated through idempotent processing, deduplication keys, and reconciliation logic.

Privacy and Security

Travel data is sensitive and internationally visible, mitigated through strong authentication, encryption, auditability, and least-privilege access.

Operational Visibility

Distributed systems are harder to troubleshoot, mitigated through tracing, metrics, logging, and proactive alerting.

Thank You

